Muhammad Ahmed Atif

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Research interests. Multi-agent reinforcement learning; cooperative AI; game theory; collective intelligence. Specifically, how the choice of welfare objective in a multi-agent system determines whether cooperation is durable when agents differ in capability.

Education

B.S. Computer Science, Habib University, Karachi — 2020 – 2024

CGPA 3.32/4.00, 131 credits. Merit scholarship covering 70% of tuition, awarded on entry and held from 2020 to 2024.
Senior Capstone (Kaavish I & II), two semesters, supervised, 6 credit hours: *"Design and Development of an Autonomous Indoor Food Delivery Robot,"* May 2024. Four-person ROS 2 build; I owned navigation and motion planning, released separately as the study published at [2]. navigation and planner code team thesis and project archive

Publications

Peer-reviewed, published

[1] **M. A. Atif** and M. S. Shaikh. "Bridging the Gap Between Theoretical and Practical Reinforcement Learning in Undergraduate Education." *2025 IEEE Frontiers in Education Conference (FIE)*, November 2025, pp. 1–9. doi:10.1109/FIE63693.2025.11328225 arXiv:2509.05689

Finding: an integrated lecture-and-lab framework on Gymnasium environments, evaluated against student surveys, instructor feedback and course performance. Gains concentrate in debugging, parameter tuning and model evaluation, where theory-first teaching leaves students weakest.

[2] S. M. Hussain, **M. A. Atif**, S. M. Daniyal, L. Ahmed, B. Memon and Q. Pasta. "Navigating the Maze Environment in ROS-2: An Experimental Comparison of Global and Local Planners for Dynamic Trajectory Planning in Mobile Robots." *2024 IEEE International Conference on Robotics and Automation in Industry (ICRAI)*, December 2024, pp. 1–7. doi:10.1109/ICRAI62391.2024.10894622

Finding: A*, Dijkstra, Regulated Pure Pursuit and the Dynamic Window Approach paired as global and local planners in a dynamic maze, scored on path length, trajectory deviation and execution time. Which pairing wins depends on the metric, so the two choices cannot be made independently.

Contribution: designed and ran the comparison and its ROS 2 harness; lead author of the public code, rosbag recordings and plots.

Under review

[3] **M. A. Atif** et al. "PPAGE: A Predator-Prey Gridworld as an Undergraduate On-Ramp to Multi-Agent Reinforcement Learning." Submitted to the *Seventeenth Symposium on Educational Advances in Artificial Intelligence (EAAI-27)*, held with AAAI-27, Montréal. Double-blind; notification November 2026.

Finding: MARL is taught almost only at graduate level because its tools assume a preparation floor they never state. An enforced extension contract lowers it: across a year five undergraduate contributors extended the testbed without any of them modifying its protected core, and three with no prior RL coursework independently implemented the same deep baseline against the interface.

[4] **M. A. Atif**, N. N. Haji, M. E. Atif and M. S. Shaikh. "Does Centralized Learning Help Embodied Agents Coordinate, and When Can a Tabular Study Tell?" Under review at *AAMAS 2027*. Preprint: arXiv:2601.17454.

Finding: with coordination measured directly, centralized Q-learning cost its agents in both roles on both grids, and its predators coordinated no more than independent learners. Its one benefit came from acting on the best joint action, which cut capture time by 42% on the small grid against faster prey but helped nowhere on the larger grid. A tabular comparison can be trusted only if it also checks state-space coverage and implements the methods alike.

Work in progress

[5] A. Nasir, F. Dossa, **M. A. Atif** and M. S. Shaikh. "A Controlled Study of Double DQN and Dueling DQN Under Cross-Environment Transfer." arXiv:2602.09810. Reviewed at IJCNN 2026 (IEEE World Congress on Computational Intelligence); revised version in preparation for *Transactions on Machine Learning Research* (TMLR).

Finding: CartPole to LunarLander under one fixed layer-wise transfer protocol and identical hyperparameters. DDQN avoids negative transfer; Dueling DQN shows it consistently, with degraded rewards and unstable optimisation. Architectural inductive bias, rather than training budget, tracks transfer robustness here.

Contribution: supervised the study end to end and edited the manuscript. The two lead authors are undergraduates I trained; the work is theirs by design.

[6] Software paper describing PPAGE, for the *Journal of Open Source Software*. In preparation.

Research Funding

Author of both proposals; each won competitive internal funding. Under Habib University rules a student cannot hold an internal grant, so Dr. M. S. Shaikh is principal investigator of record.

- *An Open-Source Multi-Agent Reinforcement Learning Testbed for Undergraduate Research.* Summer Tehqiq Research Program, Habib University, 2026. Ten-week funded programme, competitive review.
- *Multi-Agent Reinforcement Learning: A New AI Frontier.* Habib University Internal Research Grant, 2025. Funded the university's first multi-agent learning project.

Research Software and Other Selected Projects

PPAGE: Predator-Prey Archetypal Gridworld Environment — released v0.9.2, Aug 2026

Open-source testbed for controlled study of coordination, equilibrium selection and generalisation in multi-agent reinforcement learning, and the testbed the AAMAS study [4] runs on.

- A protected physics core behind typed plugins: rewards, observations and actions are swappable registries, so changing the objective under study is a configuration change and not a rewrite. This isolates the algorithm from the environment, which conventional course sequences confound.
- Eight example learning baselines behind a single two-method interface, spanning tabular Q-learning to asynchronous actor-critic, including a game-theoretic learner that is computationally distinct yet needed no change to the core.

- Reproducible by construction: one fixed seed reproduces world construction and rollouts bit for bit, so an ablation isolates exactly one changed variable.
- Apache 2.0, Python 3.10–3.12, 312 commits on `main`, six named software authors in `CITATION.cff`, documented install, quickstart and API reference.
- doi:10.5281/zenodo.21833286 (archived, all versions) github.com/UHUMALAB/PPAGE docs

Lip2Urdu: Urdu speech synthesis from lip reading, for ataxia 2022 – 2023

Lip-to-speech system for people whose loss of motor coordination affects speech and hand control alike, so they can neither speak intelligibly nor sign, and for whom no Urdu tool existed. PyTorch pipeline covering a CNN facial encoder, audio preprocessing and synthesis, following the Lip2Wav and Tacotron2 line of work. Led a three-person team, 18 of 24 commits. Second place at DURS 2024.
github.com/ProValarous/Lip2Urdu

Research Experience

Research Supervisor, Summer Tehqiq Research Program Habib University, Jun – Aug 2026

Under Dr. M. Shahid Shaikh. Funded ten-week programme to harden the MARL testbed into a releasable research and teaching tool and widen its algorithm library.

- Supervised a team of three undergraduate researchers, each owning a component, and ran the v0.9.2 release.
- Architected the layered plugin design separating environment, observation, reward and action, so that five undergraduate contributors extended the testbed across a year without modifying its core.
- Implemented joint-action learning with game theory (JAL-GT) using Correlated-Q, verified against the reference textbook's worked examples, then traced its failure to learn to a degenerate game in the experimental setup and not to the algorithm.

Lead Researcher, MARL and Emergent Social Behaviour Habib University, Jul 2025 – Mar 2026

Under Dr. M. Shahid Shaikh. No multi-agent learning group existed at the institution; the project established one. Written up as [4].

- Designed the study isolating coordination structure from function approximation, and ran the full grid: 4 pairings × 3 regimes × 10 seeds, 40,000 episodes each, with fixed seeds and full state-space enumerability.
- Designed a two-phase best-response evaluation against frozen predators, showing the trained policies remained exploitable and bounding what the comparison claims.
- Traced an apparent advantage for independent learners on the 8 × 8 grid to how the two methods break ties, and proposed checks for when a tabular comparison can be trusted.
- Built and released PPAGE; trained and supervised the undergraduate research team.

University Research Assistant, Reinforcement Learning Habib University, Sep 2024 – Jun 2025

Research-weighted appointment (approximately four-fifths research) on the gap between how reinforcement learning is taught and what students can do with it. Designed and evaluated the lecture-and-lab framework, measured its effect on practical competency, published as [1].

Teaching and Mentoring

- **Co-taught Introduction to Reinforcement Learning (300 level), Habib University, Spring 2025.** The 26-student offering evaluated in [1], run on the lecture-and-lab framework I designed.
- **Teaching Assistant, Habib University, 2022–2023.** Probability & Statistics; Programming Fundamentals.
- **President, Habib University Artificial Intelligence and Robotics Club, 2022–2023.**

Industry Experience

Research Engineer, SiliconZero, Inc. California, USA (remote), Apr 2026 – Present

Founding team. End-to-end automation of analog IC design under hard physical and electrical correctness constraints.

- Built the design pipeline: a circuit written in Python becomes, in one command and nine stages, a SkyWater sky130 layout that passes DRC and LVS, with post-layout extraction, on PADE and LAYGO2 driving KLayout, Magic, netgen, ngspice and xschem. Demonstrated end to end to a customer, who affirmed it.
- Designed the product's schematic abstraction (primitives, motifs, complete circuits) within two months of my first CMOS inverter layout.
- Led the development of the product's user interface, the part customers see: navigation, schematic and layout views, analysis plots, a schematic editor, hierarchical design representation and an artifact explorer, over a FastAPI rendering service (SPICE to schematic, GDS to layout).
- The pipeline is the execution layer for the product's planned agentic design loop.

Artificial Intelligence Intern, Folio3 Software Karachi, Jun – Jul 2022

- Built and evaluated vision models for clothing-care symbol recognition, including data curation and augmentation.

Honours and Awards

- **Second position, Invent For The Planet Global Finals**, Texas A&M University, 2023, after placing **first in the regional round**. A 48-hour design competition run at 24 universities worldwide, each host's winner advancing to College Station.
- **Second place, Dhanani School Undergraduate Research Symposium (DURS)**, Habib University, 2024, for *Lip2Urdu* (above). DURS is the school's annual juried symposium and the first of its kind in Pakistan.
- **Regional Qualifiers, International Collegiate Programming Contest 2023.**
- **Winner, Omicron II**, Habib University, 2019 (pre-university).
- **Winner, Junior Junkies Module, Student Engineering Exhibition**, Karachi Institute of Economics and Technology (KIET), 2017 (pre-university).

Skills and Languages

Programming: Python, C++, C, TypeScript, FastAPI, ROS 2, Git, LaTeX.
ML and RL: PyTorch, TensorFlow 2, Gymnasium, RLlib, Tianshou.
IC design tools: KLayout, Magic, netgen, ngspice, xschem; SkyWater sky130.
Languages: English (IELTS Academic 7.5, CEFR C1); Urdu (native); Arabic (classical; grammar and morphology); Punjabi (conversational); German (beginner).